

HR ANALYTICS DASHBOARD

1. PROJECT OVERVIEW

This project analyzes Human Resource (HR) data using Power BI, SQL, Python, and Excel. The dashboard helps organizations monitor workforce metrics, employee demographics, salary distribution, recruitment effectiveness, employee performance, and satisfaction levels. It enables HR teams to make data-driven decisions for workforce planning and organizational growth.

2. TOOLS USED

Excel	Data Cleaning & Preparation
SQL Server	Data Querying & Analysis
Python	Data Analysis & Visualization
Power BI	Interactive Dashboard Creation

3. KEY KPIS

- Total Headcount: 45
- Active Employees: 29
- Terminated Employees: 16
- Male Employees: 22
- Female Employees: 23
- Total Salary Expense: 3.1M

4. DASHBOARD INSIGHTS

- Production department has the highest employee count.
- LinkedIn is the most effective recruitment source.
- Majority of employees fall under the “Fully Meets” performance category.

- Most employees report Acceptable to High satisfaction levels.
- Production department contributes the highest salary expenditure.
- Workforce distribution is concentrated across selected states.

5. SAMPLE SQL QUERIES

TOTAL HEADCOUNT

```
SELECT COUNT(*) AS Headcount  
FROM HRData;
```

ACTIVE EMPLOYEES

```
SELECT COUNT(*) AS Active_Employees  
FROM HRData  
WHERE EmploymentStatus = 'Active';
```

TERMINATED EMPLOYEES

```
SELECT COUNT(*) AS Terminated_Employees  
FROM HRData  
WHERE EmploymentStatus = 'Terminated';
```

GENDER DISTRIBUTION

```
SELECT Gender,  
COUNT(*) AS Employee_Count  
FROM HRData  
GROUP BY Gender;
```

DEPARTMENT-WISE EMPLOYEE COUNT

```
SELECT Department,  
COUNT(*) AS Employee_Count  
FROM HRData  
GROUP BY Department  
ORDER BY Employee_Count DESC;
```

DEPARTMENT-WISE SALARY

```
SELECT Department,  
SUM(Salary) AS Total_Salary  
FROM HRData  
GROUP BY Department  
ORDER BY Total_Salary DESC;
```

RECRUITMENT SOURCE ANALYSIS

```
SELECT RecruitmentSource,  
COUNT(*) AS Total_Hires  
FROM HRData  
GROUP BY RecruitmentSource  
ORDER BY Total_Hires DESC;
```

EMPLOYEE PERFORMANCE ANALYSIS

```
SELECT PerformanceRating,  
COUNT(*) AS Employees  
FROM HRData  
GROUP BY PerformanceRating;
```

EMPLOYEE SATISFACTION ANALYSIS

```
SELECT SatisfactionLevel,  
COUNT(*) AS Employees  
FROM HRData  
GROUP BY SatisfactionLevel;
```

6. SAMPLE PYTHON ANALYSIS

IMPORT LIBRARIES

```
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt
```

LOAD DATASET

```
HR_df = pd.read_csv("HRData.csv")  
HR_df.head()
```

TOTAL HEADCOUNT

```
Headcount = HR_df['EmployeeID'].count()
print("Total Headcount:", Headcount)
```

TOTAL SALARY

```
Total_Salary = HR_df['Salary'].sum()
print("Total Salary:", Total_Salary)
```

EMPLOYEE COUNT BY DEPARTMENT

```
Dept_Count = HR_df.groupby('Department')['EmployeeID'].count()
```

```
Dept_Count.plot(kind='bar')
plt.title("Employee Count by Department")
plt.xlabel("Department")
plt.ylabel("Employees")
plt.show()
```

GENDER DISTRIBUTION

```
HR_df['Gender'].value_counts().plot(
kind='pie',
autopct='%1.1f%%'
)
plt.title("Gender Distribution")
plt.show()
```

RECRUITMENT SOURCE ANALYSIS

```
Recruitment = HR_df['RecruitmentSource'].value_counts()
```

```
Recruitment.plot(kind='barh')
plt.title("Recruitment Source Analysis")
plt.xlabel("Hires")
plt.show()
```

SALARY DISTRIBUTION

```
HR_df['Salary'].plot(kind='hist', bins=20)
```

```
plt.title("Salary Distribution")
plt.xlabel("Salary")
plt.show()
```

7. DASHBOARD OVERVIEW



8. CONCLUSION

This project successfully transformed raw HR data into actionable business insights using SQL, Python, and Power BI. The dashboard provides a comprehensive view of workforce demographics, salary expenses, employee performance, recruitment effectiveness, and satisfaction levels. It demonstrates how HR analytics can support strategic decision-making, workforce planning, and organizational growth through data-driven insights.